An Efficient Continual Learning Grasp Detection Method With Lightweight RGB-D Fusion and Automated Data Annotation for Kitchen Waste Sorting

Songyun Deng, Yaoxin Lai, Li Zhou, Ruiyuan Pei, Hai Qin, Shao Xiang, Qiaokang Liang, *Senior Member, IEEE*, Wei Sun, Dan Zhang, *Senior Member, IEEE*, and Yaonan Wang

***Abstract*—Utilizing robots for sorting and impurity removal of kitchen waste presents a challenging task. When performing such tasks, visual servo systems often require a mechanism to continuously accommodate the new tasks and environments. First, we propose a lightweight RGB-D fusion module, which efficiently fuses RGB and depth information at the channel level, achieving an over 2.1% improvement in grasping accuracy compared to traditional RGB-D fusion methods without increasing computational burden. Second, we introduce a labels automated annotation module based on detecting the degree of gripper closure. This module determines grasp success or failure by assessing the actual closure of the gripper. Finally, through these labels and manual correction, we constructed five grasping tasks to build a new continual learning dataset [kitchen waste grasp on continual learning (KWG-CL)] and proposed a novel continual learning method. By using this method, our algorithm achieves the highest average image-level grasp accuracy (77.28%) on the KWG-CL compared to state-of-the-art approaches, while exhibiting the least catastrophic forgetting. In addition, we also validated this in a real-world robot system. To the best of the authors' knowledge, this is the first application of the continual learning method in kitchen waste sorting.**

***Index Terms*—Automated data collection, continual learning, generative grasp model, kitchen waste sorting, RGB-D fusion.**

Received 17 June 2025; revised 4 November 2025; accepted 27 December 2025. Recommended by Technical Editor M. Liarokapis and Senior Editor J. Ueda. This work was supported in part by the National Natural Science Foundation of China under Grant 62073129 and Grant U21A20490, in part by the Hunan Provincial Natural Science Foundation of China under Grant 2022JJ10020, and in part by the Scientific Research Project of Hunan Education Department under Grant 21B0330. *(Corresponding author: Qiaokang Liang.)*

Songyun Deng, Yaoxin Lai, Li Zhou, Ruiyuan Pei, Hai Qin, Qiaokang Liang, Wei Sun, and Yaonan Wang are with the National Engineering Research Center for Robot Vision Perception and Control, School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China (e-mail: songyun@hnu.edu.cn; lyx03@hnu.edu.cn; tda2030@hnu.edu.cn; peiruiyuan@hnu.edu.cn; qinhai@hnu.edu.cn; qiaokang@hnu.edu.cn; wei_sun@hnu.edu.cn; yaonan@hnu.edu.cn).

Shao Xiang is with the College of Information Engineering, Zhejiang University of Technology, Hangzhou 310023, China (e-mail: shine678@zjut.edu.cn).

Dan Zhang is with the Department of Mechanical Engineering, The Hong Kong Polytechnic University, Hong Kong (e-mail: dan.zhang@polyu.edu.hk).

This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/TMECH.2025.3650331.

Digital Object Identifier 10.1109/TMECH.2025.3650331

I. Introduction

URBAN kitchen waste is a major source of solid waste and a key contributor to environmental issues, such as climate change, soil contamination, and groundwater pollution. Due to the complexity of the recycling environment, biological risks associated with manual sorting, and a persistent shortage of labor, automated sorting methods based on robotics have become the primary approach in this field [1], [2]. In the kitchen waste sorting scenario, the robot's core task is to separate recyclable and nonbiodegradable impurities from the mixed waste pile on the conveyor belt. This process relies primarily on the robot's vision servo system for rapid and precise grasping planning of the targets. Generative grasp detection methods based on the 5-D grasping rectangle are emerging as the mainstream approach in 2-D grasp detection due to their pixel-level accuracy and end-to-end, single-stage architecture. Unlike discriminative methods, they directly predict grasp attributes per pixel rather than performing object detection [3]. However, these methods often encounter challenges, such as high computational demands and low training efficiency. Some methods address the computational cost by designing lightweight modules [4], while others improve training efficiency by enhancing the network's feature extraction capabilities [5]. Nevertheless, these methods do not take into account the interrelationship between RGB and depth information. The experimental findings in [6] and [7] demonstrate that combining both RGB and depth information results in higher grasping accuracy compared to using either modality alone. Hence, developing a lightweight yet high-performance RGB-D fusion module is critical for achieving efficient grasping methods.

On the other hand, in kitchen waste sorting scenarios, challenges, such as data scarcity, difficult annotation, and the

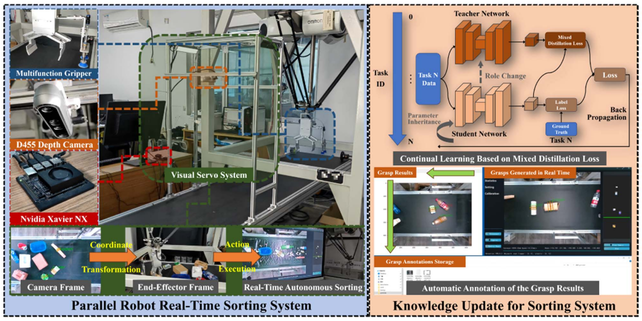

Fig. 1. Kitchen waste sorting system using our methods. It can autonomously labels 5-D grasping rectangles on RGB-D images. After manual verification and correction of the labels, the model's knowledge is continuously updated through the proposed continual learning method.

complexity and variability of sorting targets, are often encountered [8]. Especially, sorting targets often undergo significant changes in appearance due to factors, such as surface dirt accumulation and collision, and the objects to be sorted may change over time. Therefore, the sorting system must incorporate continual learning capabilities to effectively address dynamic scenarios over time. In robotics, the term continual learning refers to the ability of a robot to learn new skills without forgetting or with minimal forgetting of previously acquired knowledge [9]. Currently, research on continual learning in robotic grasping is relatively limited compared to other fields, and existing methods struggle to achieve high-accuracy grasp detection without using samples from previous tasks [10].

To solve the aforementioned issues, we make improvements in three areas: RGB-D information fusion architecture, automated dataset collection and annotation, and continual learning frameworks. Fig. 1 illustrates the system architecture and the application of our theoretical methods. Overall, our contributions are summarized as follows.

1) This work proposes an efficient RGB-D multiobject grasping detection model. Through the combination of a lightweight cross-channel fusion (LCCF) module and the lightweight advantage of PP-LiteSeg [11], this model can achieve a grasping accuracy of 98.9% on the Cornell dataset [12] with a computational cost of 3.06 GFlops, and obtain a grasping accuracy of 89.25% on the KWG2024 dataset [13] with a computational cost of 3.23 GFlops. Compared with the advanced networks, our method can achieve the most efficient grasping detection.
2) We present a method for the automated acquisition and annotation of RGB-D images and propose the first continual learning dataset for the kitchen waste sorting scenario. We employ a method to determine the success of grasping process by detecting whether the gripper can reach its limit position during closure under a constant force. This significantly reduces the human workload of data collection and annotation.
3) We propose a continual learning framework based on mixed distillation loss. This method transfers optimal student model parameters from the previous task to both student and teacher models for the current task, and the novel mixed distillation loss significantly alleviates catastrophic forgetting in student models during new task learning. Based on this method, our model can achieve the highest mean image-level grasp accuracy GA(IL) of 77.28% in the kitchen waste grasp on continual learning (KWG-CL) compared to the state-of-the-art methods in recent years.
4) We conducted a large number of sorting experiments by simulating the five scenarios of KWG-CL in the actual parallel robot sorting system. Eventually, after five tasks of experiments, the proposed method achieved the highest mean grasp success rate of 87.08% and the highest mean sorting accuracy rate of 90.95% compared to other methods.

II. Related Works

A. Real-Time Generative Grasp Detection Methods and RGB-D Fusion

Generative grasping convolutional neural network (GGCNN) [3] pioneered generative grasp detection in 2-D planar grasping by demonstrating that neural networks can decouple grasp prediction into quality, angle, and gripper width through feature maps. Subsequent approaches, such as generative residual convolutional neural network (GR-ConvNet) [6] and squeeze-and-excitation residual network (SE-ResUnet) [14], built on this by improving feature extraction or using attention mechanisms to enhance accuracy, ultimately achieving near 100% grasp success rates on the Cornell dataset [15], [16]. Nevertheless, while these methods improved the accuracy, they significantly augmented the computational cost of the models. Enabling efficient utilization of feature information is crucial for achieving a balance between high accuracy and real-time performance. Some approaches were proposed to achieve efficient feature utilization, employing architectures, such as multidimensional attention fusion [17] and depthwise separable convolutions [4]. However, these methods do not consider improving the efficiency of feature extraction through the fusion of RGB and depth information. The fusion of RGB and depth information serves as an effective feature extraction approach to achieve efficient utilization of features. Depending on whether the stage of fusing RGB and depth information occurs before or after the CNN layer, the fusion methods can be classified into early fusion and late fusion [7]. Most generative grasp detection approaches adopt early fusion strategies, where RGB and depth information are concatenated directly along the channel dimension to form four-channel RGB-D data [18]. However, early fusion fails to encompass the complementarity between the RGB information and the depth information. To solve this problem, recent studies have introduced dense fusion [7], [19], [20] or multiscale feature fusion [21], [22] that adaptively explore the correlations between these two modalities. On the other hand, some methods explore the associations by analyzing the relative spatial relationships between dual images such as multimodal embedding and convolutional modulation transformer (MCT) [23] or by leveraging information flow across channels [24]. However, most of these fusion architectures significantly increase computational burden, with only a few considering the lightweight fusion framework [7], [22].

B. Continual Learning in Generative Grasp Detection

The robot sorting system requires interaction with the real world and is limited in power and memory capacity by practical conditions. Thus, continual learning is naturally applicable to it. However, existing systems designed for kitchen waste sorting have not incorporated continual learning methods to adapt to dynamically changing scenarios [8], [13]. Besides, in the past, continual learning was mostly used in problems, such as image classification, with only a very small part related to robot grasp detection [9]. In the aspect of robot 2-D grasp, Auddy et al. [25] established the first continual learning benchmark based on the Cornell dataset, while Yang et al. [10] constructed the first continual learning benchmark based on the Jacquard dataset [26]. In these benchmarks, they mainly contrasted the distinctions of continual learning methods based on regularization, knowledge distillation, and memory replay on these datasets. Nevertheless, the continual learning version of these two datasets are not publicly accessible at present. Moreover, Liu et al. [27] proposed a knowledge distillation framework for 2-D continual learning grasping. However, this benchmark cannot be applied to the multiobject grasping issue in the conveyor belt scenario.

III. Problem Statement

A. Definition of Grasping Representation

In this article, compared with the grasp pose representation proposed by Morrison et al. [3], we add the description of the object category to adapt to the task of kitchen waste sorting [13]. Specifically, for the robot coordinate system, a grasp pose can be represented as

$$\mathbf{G}_r = (\mathbf{P}_r, q_r, \theta_r, w_r, c_r) \tag{1}$$

where $\mathbf{P}_r$ denotes the coordinates of the grasp center and can be further represented as $\mathbf{P}_r = (x_r, y_r, z_r)$. Whereas q_r, θ_r, w_r, and c_r, respectively, denote the quality, rotation angle, width, and the category of the grasp rectangle. Particularly, the quality q_r represents the probability of a successful grasp at a certain position, with the value ranging from 0 to 1. The value closer to 1 indicates a higher chance of grasp success.

Since our grasps are performed through visual servo, a grasp pose representation in an N-channel image $I \in \mathbb{R}^{N\times H\times W}$ is $\mathbf{G}_i = (\mathbf{P}_i, q_i, \theta_i, w_i, c_i)$, where $\mathbf{P}_i$ denotes the gripper's centre position (x_i, y_i), $\theta_i \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ is the angle of the gripper rotation, $w_i \in [0, w_{\max}]$ means the width of the gripper's opening, and $c_i \in [0, 1, 2, \ldots, 9]$ indicates the category of the target currently grasped by the gripper. For the execution of the grasping operation, the representation in the pixel coordinate is required to be transformed to that in the robot coordinate

$$\mathbf{G}_r = T_{rv}\left(T_{vc}T_{ci}\left(\mathbf{G}_i\right) + \Delta\mathbf{L}\right) \tag{2}$$

where T_{rv} is the transformation from conveyor belt frame to robot frame, T_{vc} is the transformation from camera frame to conveyor belt frame, T_{ci} is the transformation from image frame to camera frame, and $\Delta\mathbf{L}$ is a time-varying variable to represent the displacement of the target moving with the conveyor belt within the unit time.

B. Definition of Continual Learning Task

An advanced continual learning model needs to continuously learn new data without using the past training data. In this article, we assume that $F(\cdot)$ is a method for jointly completing the dataset by humans and the sorting system, thus

$$D_\tau = F\left(D_0\right), \tau \in \{1, 2, 3, 4, 5\} \tag{3}$$

where D_τ denotes the data of the task τ and D_0 denotes the data in KWG2024. Thus, the training processes of the five continual learning tasks can be expressed as $\mathrm{NET}_\tau = tr(\mathrm{NET}_{\tau-1}, D_\tau)$, where NET_τ denotes the network well-trained under the task τ, and** $\mathrm{tr}(\cdot)$ means the proposed continual learning training method for the task τ. Notably, D_τ can merely be utilized for training the NET_τ.

IV. Methods

A. LCCF Module: A Module for Efficient Fusion of RGB-D Information

To achieve efficient RGB-D information fusion, we propose a LCCF module, as depicted in Fig. 2. Inspired by the channels exchange mechanism in multimodal fusion with channel and spatial attention residual network (MCS-ResNet) [24], we implement cross-channel information flow across different data sources in a lightweight manner. In the first fusion stage, the input is initially divided into three distinct modalities. Each modality undergoes feature extraction through individual convolutional layers. After sequential concatenation of the extracted features, the combined tensor is processed by an SE module before being partitioned into upper and lower segments. At this stage, the features from different sources remain independently extracted with few cross-channel information. Then, both upper and lower segments undergo separate feature extraction, while the complete tensor receives additional feature extraction. At this stage, each feature layer has undergone localized information fusion, incorporating features from RGB, depth, and RGB-D sources. Subsequently, these feature layers are sequentially concatenated and passed through an SE module. Leveraging the advantages of localized feature fusion and the lightweight benefits of group convolution [28], this fusion architecture achieves performance improvement alongside reduced computational complexity compared to dense fusion [7], [20]. Specifically, with reference to the formulation by Andrew et al. [29], we express the computational cost of standard convolution as $\mathrm{FLOPs} = 2 \cdot K^2 \cdot M_F^2 \cdot C_i \cdot C_o$, where K denotes the kernel size of a convolution, M_F denotes the output feature map size, C_i is the number of input channels, and C_o is the number of output channels. We assign a, b, and c as the identifiers for each group, respectively. In our method, the output feature map sizes of the three groups are the same, the kernel sizes are the same, and there is a relationship $C_{ia} + C_{ib} = C_i$. Hence, as illustrated in the process description of Fig. 2, the computational cost of the convolutional portion subsequent to a fusion operation can

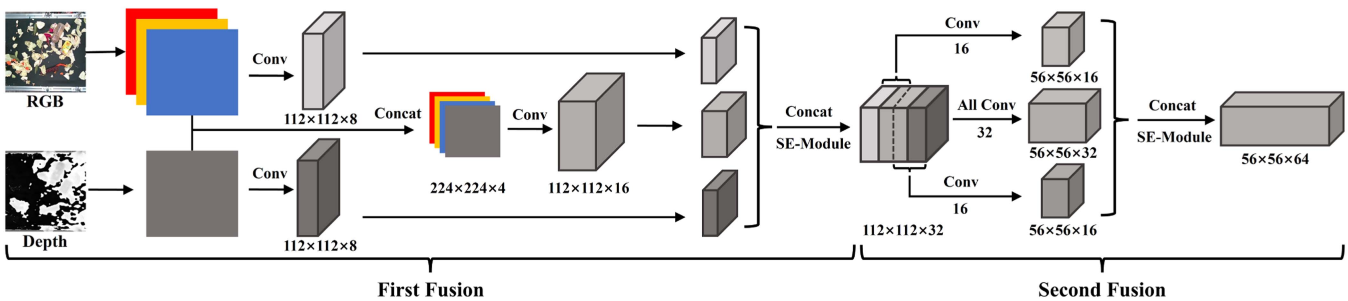

Fig. 2. Structure of LCCF module.

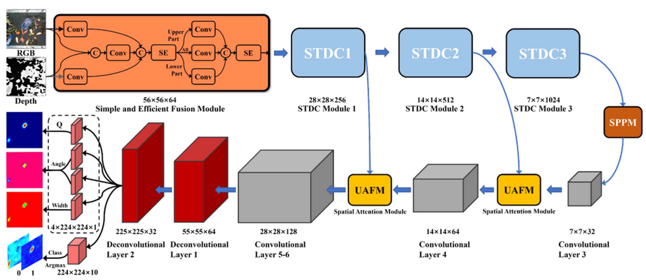

Fig. 3. Proposed architecture: PP-MixNet.

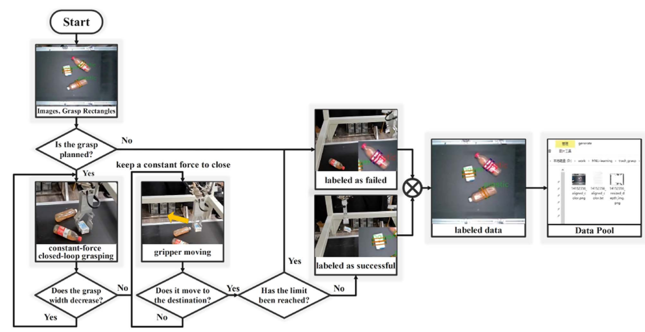

Fig. 4. Flowchart of automated data annotation for our system.

be represented as

$$\text{FLOPs}_{\text{fusion}} = 2 \cdot K^2 \cdot M_F^2 \cdot C_i \cdot \frac{3}{4} C_o. \tag{4}$$

Consequently, our method transforms the computational load to approximately three-fourths of that of the original method.

B. Network Architecture of PP-MixNet

Since the generative grasp detection network is highly similar in structure to the semantic segmentation type networks, both composed of encoders and decoders, we consider achieving efficient grasp detection by improving the advanced lightweight semantic segmentation network, PP-LiteSeg. At the front end of the network, we replace the original common convolution with the LCCF module and generate the grasp affordance map in the form of deconvolution at the back end. Thus, we named it PP-MixNet, where the details can be found in Fig. 3. Since the LCCF module we proposed has already enhanced the interaction of information from the channel dimension, we adopt the unified attention fusion module [11] using spatial attention mechanism to complement the part that the LCCF module has not focused on. Besides, the output feature map of the grasp detection task markedly differs from that of the semantic segmentation task, presenting a more substantial variation from the original image. Hence, we utilize deconvolution rather than the original upsampling at the end of the network.

C. Automated Annotation System and KWG-CL Dataset Construction

To address the problem of scarcity of dataset, we propose an automated labeling system based on our sorting system and a continual learning dataset: KWG-CL by this system. First, the KWG2024 dataset was divided into a training set and a test set at an 8:2 ratio, and the PP-MixNet was trained for 150 epochs. Subsequently, the best-performing model from the test was deployed into our sorting system. The end effector of our system operates through constant-force closed-loop grasping, where the gripper closes with a constant force. Even if the object falls after grasping, the gripper maintains closure to the limit position through constant force application. Based on this capability, we designed an automated labeling process, as illustrated in Fig. 4. In this process, objects that fail to be grasped or are not grasped in time are labeled as failed grasps if detected by PP-MixNet. Successfully grasped objects are labeled as successful grasps. Researchers manually supervise the entire automated annotation process, and no changes are made to the labels of successful and right grasps. Adjustments are only made to the error annotations, failed grasps, and undetected objects, thereby forming the KWG-CL dataset.

In this dataset, we design five continual learning tasks, which are defined as Task 1: Simple scene, Task 2: The scene with disorderly debris, Task 3: The scene with disorderly debris & black plastic film, Task 4: The scene with black plastic film & oil-free biomass impurities, and Task 5: The scene with black plastic film & oil-contaminated biomass impurities. This simulates the transition from sorting general solid waste to sorting kitchen waste with severe contamination. Specifically, starting from Task 3, we cover the conveyor belt with black plastic film. This is done to increase the complexity of the task scenario and to protect the conveyor belt from biomass impurities and oil contamination. In KWG-CL, the grasped objects include ten categories, namely, plastic bottle, plastic box, paper box, paper cup, metal bottle, glass bottle, ceramic cup, wooden

Fig. 5. Image samples of five tasks in the kitchen waste continual learning dataset: KWG-CL.

block, foam, and others, consistent with the classification in KWG2024. Specifically, Task 1 includes 209 RGB-D images, Task 2 includes 242, Task 3 includes 256, Task 4 includes 217, and Task 5 includes 240 RGB-D images. All RGB images are aligned with the depth images and cropped from the original size of 640 × 480 to 360 × 360 through postprocessing, which is consistent with the KWG2024. Finally, sample examples of the constructed KWG-CL dataset are shown in Fig. 5, and the dataset samples can be accessed online.[1]

D. Continual Learning Architecture and Loss Function

Since the kitchen waste sorting system often needs to cope with new sorting environments and new sample types, possessing the ability of continual learning has become the development focus of this mission scenario. Our modified architecture enhances the knowledge distillation framework previously applied in robotic grasping, enabling it to better retain past experience while learning new tasks. In this architecture, the network used to handle new training is called the student network, while the network trained on the old task is named the teacher network. The process of training can be seen in Fig. 1. Specifically, before the commencement of continual learning, we trained the network NET_0 in KWG2024. Subsequently, we assign the parameters of NET_0 to both the teacher network $\mathrm{NET}_1^{\text{teacher}}$ and the student network $\mathrm{NET}_1^{\text{teacher}}$ for subsequent training. As presented in Algorithm 1, in each training round, the RGB-D data undergo a preprocessing procedure to be input into the student and teacher networks. The grasp postures output by the student network are simultaneously compared with the ground truth and the soft labels produced by the teacher network, and the back propagation is conducted through the label loss and the mixed distillation loss. After the training of each task, student network with the best performance in each task is assigned to the student and teacher networks of the next task. The primary reason for the efficacy of our continual learning architecture lies in the rational configuration of the loss function and the construction of the new mixed distillation loss. Regarding the label loss, we

[1] [Online]. Available: https://github.com/HNUsong/KWG-CL

Algorithm 1: Continual Learning Process.

Require: Task Data D_τ^{train}, D_τ^{test} from $Task_\tau$
Ensure: Trained Network NET
Training NET_0 from $Task_0$
$NET_1^{\text{student}} = NET_1^{\text{teacher}} = NET_0$
for task τ in $\{Task_1 \cdots Task_\tau \cdots Task_5\}$ **do**
  $X_\tau = \text{Pre-process}(D_\tau^{train})$ $\triangleright$ $X_\tau \in \{X_1, \ldots, X_5\}$
  $\mathbf{G}_{\text{teacher}} = NET_\tau^{\text{teacher}}(X_\tau)$
  $\mathbf{G}_{\text{student}} = NET_\tau^{\text{student}}(X_\tau)$
  $\mathbf{G}_{\text{student}} \xleftarrow[NET_\tau^{\text{student}}]{learn} \{\mathbf{G}_{\text{student}} \cup \mathbf{G}_{\text{teacher}}\}$ $\triangleright$ Using label loss and mixed distillation loss
  if $\mathbf{G}_{\text{student}}$ is the best in the D_τ^{test} **and** $\tau < 5$ **then**
    $NET_{\tau+1}^{\text{student}} \xleftarrow{parameterinheritance} NET_\tau^{\text{student}}$
    $NET_{\tau+1}^{\text{teacher}} \xleftarrow{parameterinheritance} NET_\tau^{\text{student}}$
  else if $\mathbf{G}_{\text{student}}$ is the best in the D_τ^{test} **and** $\tau = 5$ **then**
    Return $NET_\tau^{\text{student}}$
  end if
end for

employ the extensively utilized Smooth L1 loss to generate the quality, angle, and width of the grasp [6]. Concerning the grasp category, since the labels are one-hot encodings, we adopt the cross-entropy loss to compute the loss of the grasp category. Specifically, the loss function of Smooth L1 is

$$L_{q/\theta/w}^{\text{label}}\left(\mathbf{G}_I^k, \mathbf{G}_A^k\right) = \begin{cases} 0.5\left(\mathbf{G}_I^k - \mathbf{G}_A^k\right)^2, \text{if} \left|\mathbf{G}_I^k - \mathbf{G}_A^k\right| \\ \left|\mathbf{G}_I^k - \mathbf{G}_A^k\right| - 0.5, \text{otherwise} \end{cases} \tag{5}$$

where k represents the index of the combination of the generated grasp and the annotated pair. Moreover, the cross-entropy loss function of the category loss is

$$L_c^{\text{label}} = -\frac{1}{k}\sum_k \sum_{m=1}^{n} C\left(c_A^{k,m}\right) \cdot \log\left(C\left(c_I^{k,m}\right)\right)$$
$$\text{where } C\left(a^{k,m}\right) = \begin{cases} 1, \text{if} a^{k,m} = m \\ 0, \text{otherwise.} \end{cases} \tag{6}$$

In this equation, m indicates the category index and n indicates the number of these categories. Therefore, the final label loss can be expressed as

$$L^{\text{label}} = L_q^{\text{label}} + L_\theta^{\text{label}} + L_c^{\text{label}}. \tag{7}$$

On the other hand, the configuration of the distillation loss is crucial for continuous learning to alleviate catastrophic forgetting. In this issue, we put forward a hybrid distillation loss to acquire the experience of soft labels from a novel perspective. Specifically, we consider the distribution of each pixel value in the feature map of the grasp quality as a Bernoulli distribution. Hence, we need to construct two new distributions for the subsequent computations

$$\begin{cases} \mathbf{S}_q^k = \log\left(\operatorname{softmax}\left(\left[\frac{1-\mathbf{G}^q_{\text{student}}k}{t}, \frac{\mathbf{G}^q_{\text{student}}k}{t}\right]\right)\right) \\ \mathbf{T}_q^k = \operatorname{softmax}\left(\left[\frac{1-\mathbf{G}^q_{\text{teacher}}k}{t}, \frac{\mathbf{G}^q_{\text{teacher}}k}{t}\right]\right) \end{cases} \tag{8}$$

where t indicates the temperature for distillation, used to soften the labels. Based on these two new distributions, a loss function regarding the quality of grasping is constructed via the Kullback-Leibler (KL) divergence

$$L_q^{\text{distillation}} = -\frac{1}{k}\sum_k \mathbf{T}_q^k \cdot \log\frac{\mathbf{T}_q^k}{\mathbf{S}_q^k}. \tag{9}$$

Referring to (6), since the category soft labels generated by the teacher network are not strict one-hot encodings, the cross-entropy should be replaced by KL divergence

$$L_c^{\text{distillation}} = -\sum_k\sum_{m=1}^{n} \mathbf{T}_q^{k,m} \cdot \log\left(\frac{\mathbf{T}_q^{k,m}}{\mathbf{S}_q^{k,m}}\right)$$
$$\text{where} \begin{cases} \mathbf{S}_c^{k,m} = \log\left(\operatorname{softmax}\left(\frac{C\left(c^{k,m}_{\text{student}}\right)}{t}\right)\right) \\ \mathbf{T}_c^{k,m} = \operatorname{softmax}\left(\frac{C\left(c^{k,m}_{\text{teacher}}\right)}{t}\right). \end{cases} \tag{10}$$

Besides, since the labels for the angle and width feature maps are not one-hot encoded, other types of loss functions must be used to fit each pixel value. Considering that the label loss for angle and width employs Smooth L1 loss to directly fit the pixel values on these feature maps, we propose using the structural similarity (SSIM) loss function [30] to measure the distance between the student network's output and the soft labels for angle and width from a human visual perspective. To be specific, our SSIM loss function is expressed as $L_{\theta/w}^{\text{distillation}}(\alpha,\beta) = 1 - \text{SSIM}_{\theta/w}(\alpha,\beta)$, where α and β represent the feature maps output by the teacher network and the student network, respectively. Thus, this loss function calculates the loss based on the similarity in luminance, contrast, and structure between the output feature map and the soft labels. Furthermore, referring to the method proposed by Douillard et al. [31], we output the feature information before it enters the deconvolution. The output feature information can be calculated in the following formula to obtain the feature loss, thereby endowing the student network with experience from the perspective of the network structure:

$$L^f\left(\mathbf{h}^s,\mathbf{h}^t\right) = L_h^f\left(\mathbf{h}^s,\mathbf{h}^t\right) + L_w^f\left(\mathbf{h}^s,\mathbf{h}^t\right)$$
$$\text{where } L_w^f\left(\mathbf{h}^s,\mathbf{h}^t\right) = \sum_{v=1}^{V}\sum_{h=1}^{H}\left\|\sum_{w=1}^{W}\mathbf{h}^t_{v,h,w} - \sum_{w=1}^{W}\mathbf{h}^s_{v,h,w}\right\|^2$$
$$\text{and } L_h^f\left(\mathbf{h}^s,\mathbf{h}^t\right) = \sum_{v=1}^{V}\sum_{w=1}^{W}\left\|\sum_{h=1}^{H}\mathbf{h}^t_{v,h,w} - \sum_{h=1}^{H}\mathbf{h}^s_{v,h,w}\right\|^2. \tag{11}$$

In this formula, $\mathbf{h}^s$ denotes the intermediate convolutional layer from student network, $\mathbf{h}^t$ denotes the intermediate convolutional layer from teacher network, and V, W, and H represent the number of channels, the width, and the height of the layer, respectively. Finally, we can obtain the mixed distillation loss

TABLE I
TRAINING ENVIRONMENT SET

Configuration Item	Cornell	KWG2024	KWG-CL
Batch Size	8	8	8
num_works	8	8	8
Optimizer	Adam	Adam	Adam
Scheduler	-	OneCycleLR	OneCycleLR
Learning Rate	0.001	-	-
Max Learning Rate	-	0.0005	0.00005
Steps Per Epoch	-	1	1
Pct_start	-	0.16	0.15
Diversity Factor	-	5.5	5.5
Final Diversity Factor	-	2000	2000
Max Epoch	50	150	60

$$L^{\text{distillation}} = \delta L_q^{\text{distillation}} + \varepsilon L_\theta^{\text{distillation}} + \varepsilon L_w^{\text{distillation}} + \eta L_c^{\text{distillation}} + \varphi L^f \tag{12}$$

where δ, η, and ε represent the weights of the quality loss function, category loss function, and the loss function using SSIM, respectively, and φ denotes the weight of the feature loss function. Thus, combining (7) and (12), the final loss function is $L = L^{\text{label}} + \lambda L^{\text{distillation}}$, where λ is used to adjust the balance between the mixed distillation loss and the label loss.

E. Evaluation Criteria

Based on previous work [6], [27], we use the Jaccard index and angle threshold to determine whether an object can be accurately grasped, as follows:

$$\begin{cases} |\theta_I - \theta_A| < 30^\circ \\ \text{IoU}\left(\mathbf{G}_I, \mathbf{G}_A\right) = \frac{\mathbf{G}_I \cap \mathbf{G}_A}{\mathbf{G}_I \cup \mathbf{G}_A} > 0.25 \end{cases} \tag{13}$$

where the generated grasp angle is denoted as θ_I and the corresponding annotation is θ_A. G_I represents the generated grasp rectangle and $\mathbf{G}_A$ stands for the corresponding ground-truth rectangle. However, in conveyor-belt-based visual servo scenarios, the task requirement of the servo system shifts from single-object grasp planning to multiobject grasp planning in a single prediction. Therefore, based on previous work [13], the evaluation metrics specific to this scenario can be divided into three submetrics: GA(IL), GA(OL), and classification accuracy of grasped targets CA. Specifically, the GA(IL) is

$$\text{GA (IL)} = M_r/M$$
$$\text{where } M_r = \sum_{i=1}^{M} P_i, P_i = \begin{cases} 1, & \text{if } N_i' = N_i \text{ and } N_i' = b_i \\ 0, & \text{if } N_{i'}' \neq N_i \text{ or } N_i' \neq b_i. \end{cases} \tag{14}$$

In this formula, N_i indicates the number of objects that can be grasped successfully in an image and N_i' stands for the number of grasps generated within an image. The overall number of all tested RGB-D images is marked as M. In one of these images, b_i represents the number of correct grasps generated. The significance of this metric is that an image is considered correct if there is exactly one valid grasp for each object, and incorrect otherwise. Besides, the second submetric, GA(OL), is defined as: $\text{GA(OL)} = \sum_{i=1}^{M} N_i' / \sum_{i=1}^{M} N_i$. This metric reflects the proportion of successfully grasped objects among all graspable objects in the dataset. The final submetric is the

TABLE II
PERFORMANCE COMPARISON OF MODELS IN CORNELL DATASET

Method	Input Size	Modality	GFLOPs	Params	Inference(FPS)	Platform	Accuracy(%)
(2020) GGCNN [3]	300 × 300	RGB-D	1.49	72K	588	PC(NVIDIA 3090)	73.9
(2020) GR-ConvNet [6], [32]	224 × 224	RGB-D	22.08	1.9M	263	PC(NVIDIA 3090)	97.7
(2022) GR-ConvNet V2 [32]	224 × 224	RGB-D	22.07	1.9M	263	PC(NVIDIA 3090)	98.8
(2022) SE-ResUNet [14], [16]	224 × 224	RGB-D	58.47	24.0M	40	PC(NVIDIA 1080Ti)	98.2
(2022) GI-Nnet [4], [15]	224 × 224	RGB-D	13.88	592K	-	PC	98.9
(2023) SKGNet [16]	288 × 288	RGB-D	259.14	33.4M	-	PC(NVIDIA TITAN RTX)	99.1
(2023) Efficient Grasp [17]	224 × 224	RGB-D	5.70	1.2M	167	PC(NVIDIA RTX 2080TI)	97.8
(2024) QQGNN1 [4]	300 × 300	RGB-D	-	238K	358	PC(NVIDIA 3090)	97.7
PP-MixNet	**224 × 224**	**RGB-D**	**3.06**	**6.7M**	**103**	**Jetson Xavier NX**	**98.9**

Bold value indicates the results obtained by the proposed method.

TABLE III
PERFORMANCE COMPARISON OF MODELS IN KWG2024 DATASET

Method	Input Size	Modality	GFLOPs	Params	Inference(FPS)	Platform	GA(IL)%	GA(OL)%	CA%
GGCNN [3]	224 × 224	RGB-D	0.85	72K	217	Jetson Xavier NX	63.40	83.62	37.45
GR-ConvNet [6]	224 × 224	RGB-D	22.24	1.9M	33	Jetson Xavier NX	85.55	93.18	93.35
GR-ConvNet V2 [32]	224 × 224	RGB-D	22.24	1.9M	32	Jetson Xavier NX	86.20	93.91	93.01
SE-ResUNet [14]	224 × 224	RGB-D	58.51	24.0M	17	Jetson Xavier NX	89.57	95.74	91.17
GI-NNet [15]	224 × 224	RGB-D	14.05	594K	40	Jetson Xavier NX	84.43	94.09	67.64
Tian *et al.* [7]	224 × 224	RGB-D	118.60	7.3M	6	Jetson Xavier NX	89.09	95.25	99.17
MCT-Grasp [23]	224 × 224	RGB-D	16.67	37.0M	19	Jetson Xavier NX	86.84	94.40	98.78
MCS-ResNet [24]	224 × 224	RGB-D	15.67	7.0M	43	Jetson Xavier NX	87.80	94.64	98.46
PP-MixNet	**224 × 224**	**RGB-D**	**3.23**	**6.7M**	**95**	**Jetson Xavier NX**	**89.25**	**95.13**	**98.02**

Bold value indicates the results obtained by the proposed method.

TABLE IV
ABLATION STUDY RESULTS OF THE LCCF MODULE

Method	Inference(FPS)	GA(IL)%	GA(OL)%	CA%
No First Fusion	95	88.44	95.07	98.40
No Second Fusion	94	88.60	95.13	97.95
No First SE	94	88.76	95.43	98.41
No Second SE	93	88.12	95.07	98.59
PP-MixNet	**95**	**89.25**	**95.13**	**98.02**

Bold value indicates the results obtained by the proposed method.

TABLE V
PERFORMANCE COMPARISON OF RGB-D FUSION MODULES

Fusion Method	Inference(FPS)	GA(IL)%	GA(OL)%	CA%
Tian et al. [7]	83	88.12	94.88	98.33
Liu et al. [22]	62	88.76	95.25	98.60
Pei et al. [24]	85	86.52	94.46	98.20
Early Fusion	101	87.14	94.64	98.14
Late Fusion	97	87.48	94.95	98.53
LCCF (Ours)	**95**	**89.25**	**95.13**	**98.02**

Bold value indicates the results obtained by the proposed method.

classification accuracy of the grasped object categories, defined as $\mathrm{CA} = \sum_{i=1}^{M} C_i / \sum_{i=1}^{M} b_i$, where C_i represents the amount of grasps that are categorized correctly among the correct grasps in an image. This metric reflects the proportion of correctly identified categories among all successful grasps. To provide a comprehensive assessment of grasping performance, we will focus on comparing with GA(IL) in the subsequent experiments of datasets, while focus on comparing with GA(OL) in the actual robotic experiments.

On the other hand, we also need to evaluate the performance of these three metrics in the continual learning scenario. For any metric E, we can obtain its average value for the jth task, considering all previous tasks

$$\overline{\mathrm{E}}_{\tau} = \frac{1}{\tau} \sum_{j=0}^{5} \mathrm{E}_{\tau,j}. \tag{15}$$

Moreover, the backward transfer (BF) rate helps calculate the model's forgetting degree of task $j-1$ during task j [10]. A lower value indicates better performance in overcoming forgetting. Similarly, the backward transfer rate for a metric E at task j can be expressed as

$$\mathrm{BF}_{\tau}^{\mathrm{E}} = \frac{1}{\tau} \sum_{j=0}^{\tau} (\mathrm{E}_{j,j} - \mathrm{E}_{\tau,j}), \text{where } \tau \neq 0,\ j < \tau. \tag{16}$$

V. EXPERIMENTS AND RESULTS

A. Experimental Setting

Models training in this article is carried out on a server platform furnished with a 13th Gen Intel Core i9-13900 K CPU and an NVIDIA GeForce RTX 4090 GPU. These models are constructed and trained by using PyTorch, and after being accelerated with TensorRT. Subsequently, they are deployed on the Jetson Xavier NX platform as FP32 data. Our experiments are divided into three parts: validating the model's performance (see Section V-B), assessing the model's continual learning capabilities (see Section V-C), and evaluating the performance of the continual learning model in the real robotic scenario (see Section V-D). In the first experiment, following the sets in methods [13] and [14], both datasets are split into training and testing sets at 8:2 ratio. The specific experimental settings are detailed in Table I. Furthermore, for the two datasets, we enhance the input images with random rotation, random cropping, scaling, and random variations in lighting intensity. Based on the previous work [3] and our experience, the size of the Gaussian kernel for postprocessing was set as 5 on KWG2024 and 3 on Cornell. In the second experiment, following the sets in methods [10], [25], and [27], the dataset for each task was split into training and testing sets at 8:2 ratio, and the best model, validated by the metric (14) and (15), is transferred to the next task. The detailed experimental settings are detailed in Table I. In addition, the

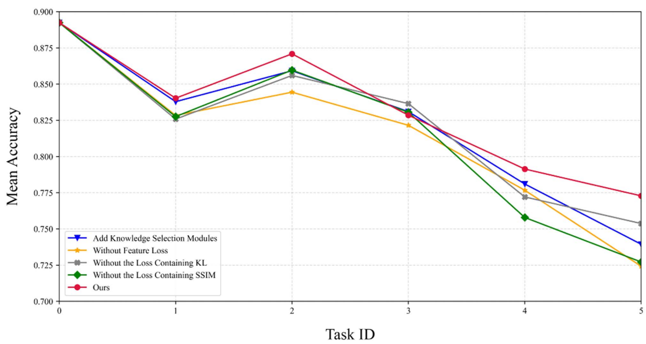

Fig. 6. Ablation study results of the proposed continual learning model.

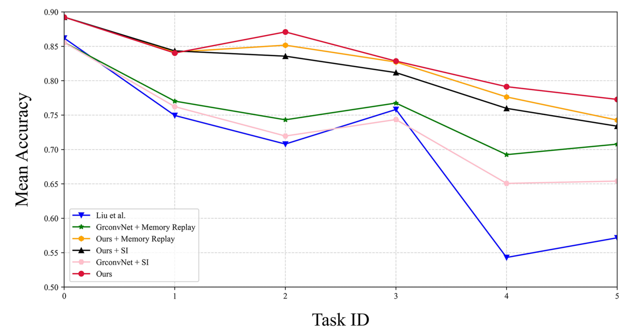

Fig. 7. Comparative experiments of mean accuracy on the KWG-CL.

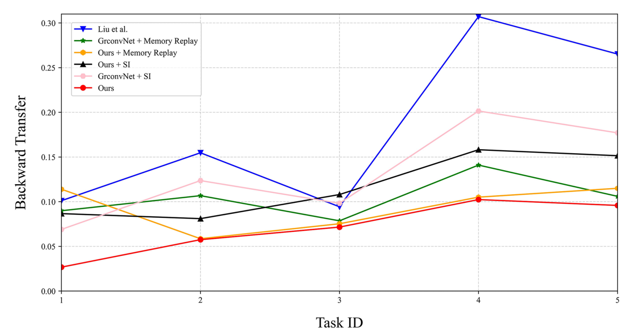

Fig. 8. Comparative experiments of backward transfer on the KWG-CL.

weight coefficients for the continual learning loss function are set as follows: $\delta = 0.001, \varphi = 0.001, \eta = 0.1, \varepsilon = 0.1, \lambda = 0.1$, and the distillation temperature $t = 3$. In the final experiment, the high-performing model from the second experiment was deployed in our real robotic sorting system for testing.

B. Experiments on PP-MixNet

1) Experiments on Cornell Dataset: In accordance with the configurations stated in Section V-A and the (13), we assessed the performance of PP-MixNet on the Cornell dataset and made a comparison with the advanced generative models in recent years. To ensure the training under conditions similar to other methods, networks in the test were trained for 50 epochs consisting of 1000 batches. Moreover, the preprocessing and postprocessing settings for the input data followed previous work [3]. To ensure the reproducibility of the experiments, we conducted the experiments using the public codebase provided by Kumra et al. [6]. The final comparison results are presented in Table II. Bold values in Tables II–VII indicate the results obtained by the proposed method.. It shows that, compared to networks with similar computational cost, our method achieves higher grasping accuracy, while compared to networks with similar grasping accuracy, our method significantly reduces computational cost. The inference speed of the model tested on Jetson Xavier NX provides further confirmation of this.

2) Experiments on KWG2024 Dataset: In this experiment, we selected models with available original code or sufficient details provided in the corresponding papers from the Cornell experiments as test candidates. Moreover, we also tested recently developed networks with efficient RGB-D fusion capabilities. We added a classification head to all networks consistent with that in Fig. 3 to generate the CA metric. For networks with unavailable code, we referred to the original papers and implemented them in Pytorch. After training, all networks were deployed in TensorRT format and tested for inference speed on the Jetson Xavier NX. Since the $\mathrm{GA(IL)}$ metric most comprehensively evaluates grasping performance in conveyor belt scenarios, the selection of the optimal model was primarily based on the epoch where $\mathrm{GA(IL)}$ performed best, and the corresponding $\mathrm{GA(OL)}$ and CA metrics were used as final results. Table III shows the performance of these networks on the KWG2024 dataset. Compared to recent networks without RGB-D fusion module, our proposed method achieves a 0.32% lower $\mathrm{GA(IL)}$ than the slowest SE-ResUNet, while its computational complexity is only 5.5% of that network's. In contrast, compared to the fastest GGCNN, our network outperforms it by 25.85% in $\mathrm{GA(IL)}$. Compared to recent networks with RGB-D fusion module, our proposed method achieves a 0.16% higher $\mathrm{GA(IL)}$ than the slowest network (Tian et al.[7]), and its computational complexity is only 2.7% of that network's. In contrast, compared to the fastest MCS-ResNet, our network outperforms it by 1.45% in $\mathrm{GA(IL)}$ and 52 FPS in inference speed. This demonstrates that our network offers higher efficiency compared to state-of-the-art methods, effectively balancing grasping accuracy and inference time.

3) Ablation Experiments: Given the limited size of the Cornell dataset, we selected the KWG2024 dataset for ablation experiments. In this process, we removed each component individually within the LCCF module, including the first fusion stage, the second fusion stage, the first SE block, and the second SE block, to evaluate the effectiveness of different structural configurations in enhancing model performance. The removed components are replaced with a convolutional layer. The experimental results are summarized in Table IV. The results demonstrate that any component contributes to performance improvement. Moreover, when the model contains only a single fusion stage, source-differentiated fusion (no second fusion) demonstrates superior performance enhancement compared to source-undifferentiated fusion (no first fusion). In addition, we replaced the LCCF module with recently developed efficient RGB-D fusion modules to validate the suitability of the LCCF module for PP-LiteSeg. The experimental results are indicated in Table V. As shown in the results, the network equipped with the LCCF module achieves the highest $\mathrm{GA(IL)}$ metric.

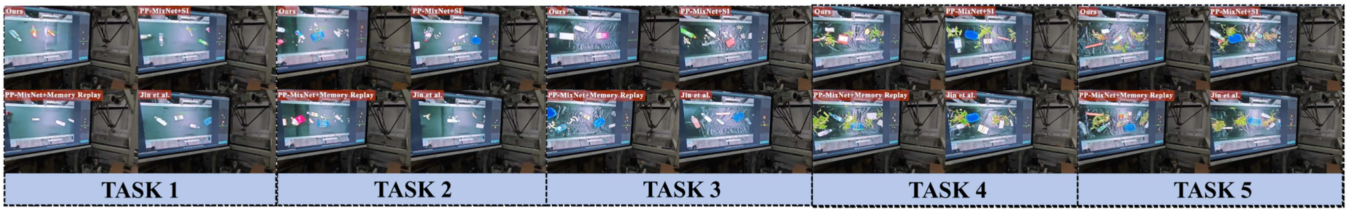

Fig. 9. Comparative experiments in the actual robot sorting system. These experiments simulated the scenarios of five tasks on KWG-CL and examined the sorting capabilities of different models for 48 samples to be grasped in these five scenarios.

Theoretically, the LCCF module reduces the computational complexity and parameters compared to early and late fusion. However, when deployed on practical platforms, it yields lower FPS. We think the reason is related to computational optimization in Pytorch and TensorRT.

C. Experiments of Continual Learning Methods on KWG-CL

1) Ablation Study: To validate the effectiveness of our continual learning method, we removed components, such as the loss function containing SSIM, the loss function containing KL divergence, the feature loss, and add the knowledge selection module proposed by Liu et al. [27]. For each removed component, the corresponding component from the label loss is added as a replacement. Notably, the inclusion of the knowledge selection modules in this ablation experiment was conducted as an additional experiment, which will be discussed in more detail in Section VI. The results of the ablation experiment are presented in Fig 6. In this figure, the vertical axis represents the model's mean $\mathrm{GA(IL)}$, which can be calculated by (14) and (15), while the horizontal axis represents the task number. It shows that removing any component leads to a significant decline in the model's performance. Specifically, removing the loss containing SSIM or the feature loss causes a considerable drop in performance, while removing the loss containing KL divergence results in a more minor decrease. This indicates that the loss containing SSIM and feature loss enable the student network to inherit more knowledge from the teacher network.

2) Comparative Experiments With Other Methods: We compared our method with other recent methods of a similar nature. These methods are selected from regularization-based methods (synaptic intelligence (SI) [25], [33]), memory replay-based methods [8], [9], and knowledge distillation-based methods [27], which represent the main research directions in this field. Moreover, since GR-ConvNet is currently the most widely used generative network in the 2-D grasping detection domain, we combined it with the SI method and the memory replay method for a comprehensive comparison. The experimental results are shown in Figs. 7 and 8, where Fig. 7 shows the mean $\mathrm{GA(IL)}$ change using (15), while Fig. 8 illustrates the backward transfer change of the $\mathrm{GA(IL)}$ using (16). The results show that our method outperforms existing methods in these continual learning tasks. Even when compared to memory replay methods (15 images for each task [10]) that utilize data from previous tasks, our method still maintains a significant advantage. From the network architecture perspective, our PP-MixNet is more suitable for continual learning tasks than GR-ConvNet, and PP-MixNet exhibits less forgetting than GR-ConvNet when using the same continual learning method. Finally, after training on five tasks, our method achieves a mean $\mathrm{GA(IL)}$ of 77.28%. The best-performing model from each continual learning method are selected for the robot system experiment in Section V-D.

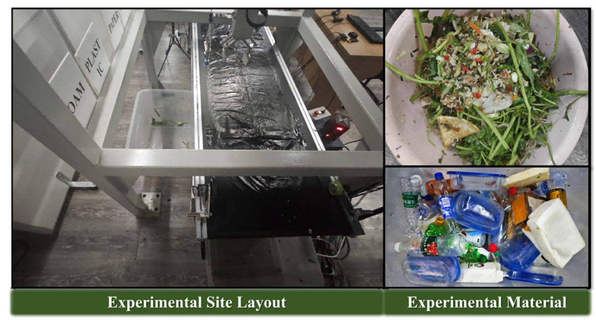

Fig. 10. Experimental environment and the materials for real-world tests.

D. Experiments on Robotic Platform

In this experiment, we simulated five scenarios from KWG-CL to evaluate the sorting performance of different methods in real-world conditions. In experimental setup, materials are shown in Fig. 10, and the materials were collected from actual kitchen waste. Moreover, a total of 48 samples were used for sorting in each scenario, and the sorting system processes them to calculate the grasp success rate and sorting accuracy [13]. The experimental procedure is illustrated in Fig. 9. Although we ensured consistency in the samples across tasks, some may undergo deformation or appearance changes during the sorting process. Furthermore, we selected the best-performing models from each continual learning method to evaluate their performance in the real-world robotic sorting system. Owing to the limitations of the actual space and the diversity of graspable objects, we restricted our tests to four categories, considering wooden blocks as paper, namely, plastic, paper, metal, and foam, to calculate the sorting accuracy rate. In addition, the experimental materials and samples were randomly positioned on the conveyor belt by the experimenters, and the robot grasped them in the sequence of their arrival. Since our robot is equipped with both a gripper and a suction cup [8], when the predicted grasp width exceeds the maximum opening distance of the gripper,

TABLE VI
RESULT OF GRASP SUCCESS RATE IN THE ACTUAL SORTING SYSTEM

Methods	Grasp Success Rate (%)					
	TASK 1	TASK 2	TASK 3	TASK 4	TASK 5	MEAN
Liu et al. [27]	77.08%(37/48)	64.58%(31/48)	62.5%(30/48)	70.83%(34/48)	62.50%(30/48)	67.50%
PP-MixNet+SI	85.42%(41/48)	83.33%(40/48)	72.92%(35/48)	79.17%(38/48)	77.08%(37/48)	79.58%
PP-MixNet+Memory Replay	81.25%(39/48)	81.25%(39/48)	**87.50%(42/48)**	75.00%(36/48)	72.92%(35/48)	79.58%
Ours	**93.75%(45/48)**	**91.67%(44/48)**	83.33%(40/48)	**85.42%(41/48)**	**81.25%(39/48)**	**87.08%**

Bold value indicates the results obtained by the proposed method.

TABLE VII
RESULT OF SORTING ACCURACY RATE IN THE ACTUAL SORTING SYSTEM

Methods	Sorting Accuracy Rate (%)					
	TASK 1	TASK 2	TASK 3	TASK 4	TASK 5	MEAN
Liu *et al.*	86.49%(32/37)	80.65%(25/31)	76.67%(23/30)	76.47%(26/34)	73.33%(22/30)	79.01%
PP-MixNet+SI	90.24%(37/41)	**92.50%(37/40)**	**91.43%(32/35)**	81.58%(31/38)	86.49%(32/37)	88.48%
PP-MixNet+Memory Replay	89.74%(35/39)	87.18%(34/39)	90.48%(38/42)	88.89%(32/36)	91.43%(32/35)	89.53%
Ours	**91.11%(41/45)**	88.64%(39/44)	90.00%(36/40)	**92.68%(38/41)**	**92.31%(36/39)**	**90.91%**

Bold value indicates the results obtained by the proposed method.

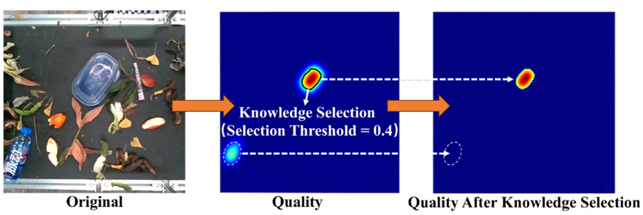

Fig. 11. Case where the knowledge selection module reduces the performance of continual learning.

the robot will use the suction cup to pick up the object based on the center of the predicted grasp rectangle. The results in our sorting system are shown in Tables VI and VII, where Table VI presents the sorting success rate of different methods across the five tasks, while Table VII shows the sorting accuracy of these methods. The final results show that our method outperforms other methods with the highest average sorting success rate and the highest average sorting accuracy. Specifically, our method exceeds the second-best method (memory replay-based and regularization-based methods) by 7.5% in sorting accuracy. Moreover, it outperforms the second-best method (memory replay-based method) by 1.38% in sorting success rate. Due to the significantly higher sorting success rate compared to the memory replay-based method, the number of correctly classified samples is also much higher. In terms of memory retention, our method achieves the highest sorting success rate of 93.75% in Task 1 of the real-world scenario, even after completing five tasks in the continual learning. Besides, the sorting success rate in Task 1 is also the highest (91.11%). This demonstrates that our method not only exhibits strong robustness but also has a strong ability to mitigate forgetting.

VI. DISCUSSION AND CONCLUSION

This article presents a generative grasp detection model, PP-MixNet, which can fuse RGB and depth information efficiently, and establishes a continual learning method for kitchen waste sorting. Our approach leverages knowledge distillation to adapt to new tasks while addressing the issue of catastrophic forgetting. Compared to similar methods [27], our approach achieves significant improvements. However, we also consider why their proposed continual learning framework performs poorly on the KWG-CL dataset. By analyzing the incorrect grasps produced by their method [27], it fails to generate grasps near both the center and the edges of the image in multiobject grasping detection. This issue arises because their knowledge selection module filters out grasps with low-quality values. We incorporated the knowledge selection module into our method during the ablation study, and the knowledge selection threshold was set to the best value among the values tested. As shown in the results of Fig. 6, the inclusion of the knowledge selection module led to a significant drop in our model's continual learning performance. This indicates that the knowledge selection module contributes to performance degradation in our tasks. By analyzing error cases, we found that while the knowledge selection module reduces noise in some instances, it tends to filter out grasps with low-quality values in many others, as illustrated in Fig. 11. On the other hand, although we have made some effective attempts in automating the dataset construction, the process still requires human intervention. Moreover, for different tasks, our method still requires manual tuning of some hyperparameters, which limits the model's ability to adapt to new tasks. Finally, integrating our automated dataset annotation with the continual learning method to achieve fully automated updates of the robot's grasping experience remains an open challenge. In future research, we will attempt to address these issues and further enhance our system's ability to adapt to greasy and cluttered scenarios in kitchen waste sorting. Furthermore, our method demonstrates strong generalization ability when test on datasets for different grasp tasks. We will further explore and discuss its generalization ability in the future.

REFERENCES

[1] Z. X. Hoy, K. S. Woon, W. C. Chin, Y. Van Fan, and S. J. Yoo, "Curbing global solid waste emissions toward net-zero warming futures," *Sci.*, vol. 382, no. 6672, pp. 797–800, 2023.

[2] A. M. Madsen et al., "Review of biological risks associated with the collection of municipal wastes," *Sci. Total Environ.*, vol. 791, Oct. 2021, Art. no. 148287.

[3] D. Morrison, P. Corke, and J. Leitner, "Learning robust, real-time, reactive robotic grasping," *Int. J. Rob. Res.*, vol. 39, no. 2-3, pp. 183–201, 2020.

[4] K. Fu and X. Dang, "Light-weight convolutional neural networks for generative robotic grasping," *IEEE Trans. Ind. Informat.*, vol. 20, no. 4, pp. 6696–6707, Apr. 2024.

[5] Z. Zhou, X. Zhu, and Q. Cao, "AAGDN: Attention-augmented grasp detection network based on coordinate attention and effective feature fusion method," *IEEE Rob. Autom. Lett.*, vol. 8, no. 6, pp. 3462–3469, Jun. 2023.

[6] S. Kumra, S. Joshi, and F. Sahin, "Antipodal robotic grasping using generative residual convolutional neural network," in *Proc. 2020 IEEE/RSJ IROS*, 2020, pp. 9626–9633.

[7] H. Tian, K. Song, S. Li, S. Ma, and Y. Yan, "Lightweight pixel-wise generative robot grasping detection based on RGB-D dense fusion," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.

[8] H. Qin et al., "Active Learning-DETR: Cost-effective object detection for kitchen waste," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.

[9] T. Lesort, V. Lomonaco, A. Stoian, D. Maltoni, D. Filliat, and N. Dıáz-Rodrıǵuez, "Continual learning for robotics: Definition, framework, learning strategies, opportunities and challenges," *Inf. Fusion*, vol. 58, pp. 52–68, 2020.

[10] R. Yang, M. Grard, E. Dellandréa, and L. Chen, "When continual learning meets robotic grasp detection: A novel benchmark on the Jacquard dataset," in *Proc. 18th Int. Joint Conf. Comput. Vis.*, Lisbon, Portugal, Feb. 2023, pp. 703–710.

[11] J. Peng et al., "PP-LiteSeg: A superior real-time semantic segmentation model," 2022, *arxiv:2204.02681*.

[12] Y. Jiang, S. Moseson, and A. Saxena, "Efficient grasping from RGBD images: Learning using a new rectangle representation," in *Proc. IEEE Int. Conf. Robot. Autom*., 2011, pp. 3304–3311.

[13] S. Deng, R. Pei, L. Zhou, H. Qin, W. Sun, and Q. Liang, "An efficient generative intelligent multiobjective grasping model for kitchen waste sorting," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–10, 2025.

[14] S. Yu, D.-H. Zhai, Y. Xia, H. Wu, and J. Liao, "SE-ResUNet: A novel robotic grasp detection method," *IEEE Rob. Autom. Lett.*, vol. 7, no. 2, pp. 5238–5245, Apr. 2022.

[15] P. Shukla, N. Pramanik, D. Mehta, and G. C. Nandi, "Generative model based robotic grasp pose prediction with limited dataset," *Appl. Intell.*, vol. 52, Jan. 2022, pp. 9952–9966.

[16] S. Yu, D.-H. Zhai, and Y. Xia, "SKGNet: Robotic grasp detection with selective kernel convolution," *IEEE Trans. Autom. Sci. Eng.*, vol. 20, no. 4, pp. 2241–2252, Oct. 2023.

[17] H. Cao, G. Chen, Z. Li, Q. Feng, J. Lin, and A. Knoll, "Efficient grasp detection network with Gaussian-based grasp representation for robotic manipulation," *IEEE/ASME Trans. Mechatron.*, vol. 28, no. 3, pp. 1384–1394, Jun. 2023.

[18] M. A. Lee et al., "Making sense of vision and touch: Self-supervised learning of multimodal representations for contact-rich tasks," in *Proc. Int. Conf. Robot. Automat.*, 2019, pp. 8943–8950.

[19] Y. Song, J. Wen, D. Liu, and C. Yu, "Deep robotic grasping prediction with hierarchical RGB-D fusion," *Int. J. Control Autom. Syst.*, vol. 20, no. 1, pp. 243–254, Jan. 2022.

[20] L. Tong and K. Qian, "HFNet: High-precision robotic grasp detection in unstructured environments using hierarchical RGB-D feature fusion and fine-grained pose alignment," *Measurement*, vol. 253, 2025, Art. no. 117775.

[21] L. Tong, K. Song, H. Tian, Y. Man, Y. Yan, and Q. Meng, "A novel RGB-D cross-background robot grasp detection dataset and background-adaptive grasping network," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.

[22] D. Liu and X. Feng, "A real-time grasp detection network based on multi-scale RGB-D fusion," in *Proc. Int. Joint Conf. Neural Netw.*, 2024, pp. 1–8.

[23] G. Yang, T. Jia, Y. Liu, Z. Liu, K. Zhang, and Z. Du, "MCT-Grasp: A novel grasp detection using multimodal embedding and convolutional modulation transformer," *IEEE Sens. J.*, vol. 24, no. 23, pp. 39206–39217, Dec. 2024.

[24] R. Pei, S. Deng, L. Zhou, H. Qin, and Q. Liang, "MCS-ResNet: A generative robot grasping network based on RGB-D fusion," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025.

[25] S. Auddy, J. Hollenstein, M. Saveriano, A. Rodrıǵuez-Sánchez, and J. Piater, "Continual learning benchmarks for antipodal grasping," in *Proc. Aust. Robot. Workshop*,Jun. 2022, pp. 1–4.

[26] A. Depierre, E. Delland'rea, and L. Chen, "Jacquard: A large scaledataset for robotic grasp detection," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst*., 2018, pp. 3511–3516.

[27] J. Liu, J. Xie, S. Huang, C. Wang, and F. Zhou, "Continual learning for robotic grasping detection with knowledge transferring," *IEEE Trans. Ind. Electron.*, vol. 71, no. 9, pp. 11019–11027, Sep. 2024.

[28] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM.*, vol. 60, no. 6, pp. 84–90, 2017.

[29] M. Andrew et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications,"2017, *arXiv:1704.04861*.

[30] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image. Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[31] A. Douillard, M. Cord, C. Ollion, T. Robert, and E. Valle, "PODNet: Pooled outputs distillation for small-tasks incremental learning," in *Comput. Vis.–ECCV*, 2020, pp. 86–102.

[32] S. Kumra, S. Joshi, and F. Sahin, "GR-ConvNet v2: A real-time multi-grasp detection network for robotic grasping," *Sensors*, vol. 22, no. 16, p. 6208, 2022.

[33] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 3987–3995.

Songyun Deng received the bachelor's degree in electronic information engineering from Central South University of Forestry and Technology and the degree of bachelor of science in Electronic Engineering with first class honors from Bangor University, Changsha, China, in 2020. He is currently pursuing a Ph.D. degree in control science and engineering with the School of Artificial Intelligence and Robotics, Hunan University, Changsha, China. His research interests include machine vision, plant diseases recognition, and robot visual servo.

Yaoxin Lai received the bachelor's degree in automation in 2024 from Hunan University, Changsha, China, where he is currently working toward a master's degree in control science and engineering with the School of Artificial Intelligence and Robotics.

His research interests include robotic grasping and continual learning.

Li Zhou received the bachelor's degree in automation from the East China University of Technology, Nanchang, China, in 2019, where he is currently working toward the master's degree in electronic information with the College of Electrical and Information Engineering, Hunan University, Changsha, China.

His research interests include machine vision and robotics system.

Ruiyuan Pei received the bachelor's degree in automation in 2023 from Hunan University, Changsha, China, where he is currently working toward the master's degree in control science and engineering with the School of Artificial Intelligence and Robotics.

His research interests include grasp detection and robotics system.

Hai Qin received the M.S degree in control science and engineering from the College of Information Engineering, Xiangtan University, Xiangtan, China, in 2015, and the Ph.D. degree in electrical engineering from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2023.

His research interests include intelligent robotic perception, computer vision, and machine learning.

Shao Xiang received the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Surveying, Mapping and Remote Sensing Information Engineering, Wuhan University, Wuhan, China, in 2024.

His research interests include artificial intelligence, pattern recognition, and remote sensing image processing.

Qiaokang Liang (Senior Member, IEEE) received the Ph.D. degree in control science and engineering from the University of Science and Technology of China, Hefei, China, in 2011.

He is currently a Professor with the School of Artificial Intelligence and Robotics, Hunan University, Changsha, China, where he is also the Vice Director of the National Engineering Research Center of Robot Vision Perception and Control. His research interests include robotic perceptions and application of artificial intelligence technology in perceptions.

Wei Sun received the B.S. degree in in automation, the M.S. degree in control science and engineering, and the Ph.D. degree in control science and engineering from the Department of Automation Engineering, Hunan University, Changsha, China, in 1997, 1999, and 2003, respectively.

He is currently a Professor with Hunan University and the Academic Leader of the National Engineering Research Center of Robot Visual Perception and Control Technology. His research interests include computer vision, robotics, neuralnetworks, and intelligent control, with more than 80 publications in these areas.

Dan Zhang (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from Laval University, Quebec City, QC, Canada, in June 2000.

He is a Chair Professor of Intelligent Robotics and Automation, Hong Kong Polytechnic University, Hong Kong. His research interests include robotics and mechatronics, and high-performance parallel robotic machine development.

Dr. Zhang is a Fellow of the Canadian Academy of Engineering, the American Society of Mechanical Engineers (ASME), and a Senior Member of the SME.

Yaonan Wang received the B.S. degree in computer engineering from the East China University of Science and Technology, Fuzhou, China, in 1981, and the M.S. and Ph.D. degrees in electrical engineering from Hunan University, Changsha, China, in 1990 and 1994, respectively.

Since 1995, he has been a Professor with Hunan University. His research interests include intelligent control and information processing, robot control, industrial process control, and image processing.

Dr. Wang is a Fellow of the Chinese Academy of Engineering.